

No. of Functional Features Included in the Solution

Date	29 September 2026
Team ID	SWTID-2026-7019
Project Name	EduGenie - AI Learning Assistant
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	AI Question & Answer	Student types any question and receives an instant AI-generated answer from Gemini.	qna.py GET /qa	Done	0.75
2	Concept Explanation	Explains a topic in simple, school-level language; uses a local Flan-T5 model when available and falls back to Gemini.	explanation_module.py POST /explain/	Done	0.75
3	Text Summarization	Condenses a pasted passage into a short summary in plain language.	summary_module.py POST /summarize/	Done	0.50
4	Quiz Generation	Creates 3 multiple-choice questions (4 options, 1 answer) as JSON from a topic or passage.	quiz_module.py POST /quiz	Done	0.75
5	Interactive Quiz & Scoring	Clickable options show correct/wrong instantly and a final score with feedback.	templates/index.html (renderQuiz, checkAnswer)	Done	0.50
6	Learning Path Recommendations	Suggests an ordered beginner/intermediate/advanced path with topics and resources.	learning_path.py GET /learn/recommendations	Done	0.75
7	Single-Page Web Interface	One page with five tool cards, loading states, toast messages and Enter-key support.	templates/index.html static/style.css	Done	0.50
8	Configuration, Validation & Error Handling	API key and models read from .env; missing-key guard, empty-input validation and JSON error responses.	main.py .env	Done	0.50
				Total	5.00

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6 (Q&A, Explanation, Summarization, Quiz Generation, Learning Path, Web Interface)
Additional / Nice-to-Have Features	2 (Interactive Quiz & Scoring, Configuration & Error Handling)
Features Tested & Verified	2 of 8 verified by automated smoke test (web page load; request validation and API-key guard). The 6 AI-dependent features need a manual run with a live Gemini key.

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Single-page web interface; interactive quiz with instant feedback and score
2	Backend / Logic	4	Explanation with local-model fallback; summarization; quiz generation (JSON parsing); learning paths
3	Database / Storage	0	None - the app is stateless; configuration is kept in .env
4	API / Integration	1	Gemini-powered Q&A endpoint (/qa); google-genai SDK used by all AI modules
5	Security / Authentication	1	API-key check, input validation and safe JSON error responses